

# SaakhScore

Saakh (साख) — a trader's earned creditworthiness. An explainable MSME Financial Health Card that proves it, built for IDBI Innovate 2026.

|                   |                                              |
|-------------------|----------------------------------------------|
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| Problem Statement | Problem Statement 3 — Financial Health Score |

## 02 BRIEF ABOUT THE IDEA

# Consented alternate data, turned into a decision a bank can defend.

### 01 What it is.

An explainable MSME Financial Health Card that scores businesses from consented alternate data.

### 03 Inputs.

GST filings, UPI transaction trails, Account Aggregator bank statements, EPFO records.

### 02 Who it's for.

New-to-Credit (NTC) and New-to-Bank (NTB) MSMEs that traditional bureau-based underwriting rejects outright.

### 04 Outputs.

A 0–100 score, an A–E grade, an indicative credit limit, and a plain-language improvement plan.

# Explainability and auditability, by default.

## DIFFERENTIATION

### Beyond a number

Most alternate-data scoring tools return a score. SaakhScore returns a decision a bank can defend to a regulator and a borrower can act on.

## APPROACH

### Deterministic core, generative shell

No LLM ever sits on the decision path. Independent, cross-checkable scoring engines run side by side; generative text only narrates a decision the deterministic core already made and logged.

## USP

### The regulatory gap

Of 127 AI-using regulated entities surveyed, only 15 use SHAP/LIME explainability and only 18 keep audit logs (RBI). SaakhScore ships both by default.

# What the health card actually does.

## 5-Pillar Health Score

Liquidity, Discipline, Momentum, Leverage, Digital Footprint → 0–100 score, A–E grade, indicative limit.

## Explainable ML + Shapley

A trained model with exact per-feature Shapley attribution behind every decision — no black box.

## Traditional vs. Alternate Verdict

Side-by-side comparison showing the rejected-to-approved reversal for thin-file businesses.

## Improvement Plan + Audit Log

A concrete action plan for borrowers, EWS-style monitoring signals, a counterparty-concentration limit guardrail, a deterministic underwriter next-best-action, and an append-only audit trail for every decision made.

## From consented data to an auditable decision.

01

### Data intake

AA statements, GST, UPI,  
EPFO



02

### Dual scoring

Rule engine + ML model, in  
parallel



03

### Verdict & explain

Traditional vs. alternate,  
Shapley reasons



04

### Memo, audit, hand-off

Memo, audit log, OCEN-  
aligned offer

## The rejected → approved moment.

The NTC hero (Shree Ganesh Textiles) has zero bureau history — declined instantly by a traditional process, approved on alternate data by SaakhScore.

TRADITIONAL (BUREAU-ONLY)

**Rejected**

No credit bureau history on file (New-to-Credit / New-to-Bank).

SAAKHSCORE (ALTERNATE DATA)

**Approved**

Grade A · Score 86/100 · Indicative limit ₹27,00,000.

# One service. Four inspectable control planes.

## Operational Edge

Request IDs, body ceilings, readiness probes, security headers, rate controls, and a non-root container.

## Decision + Evidence

Five transparent pillars, calibrated PD, exact TreeSHAP attribution, policy checks, and reason codes.

## Temporal Validation

Dated outcomes, 365-day maturity, chronological development/calibration/OOT, PSI, KS, AUC, and Gini.

## Promotion + Audit

Pseudonymous hash-chain evidence and fail-closed pilot startup until model, OOT, identity, and storage pass.

# A reproducible, inspectable open stack.

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Backend         | Python, FastAPI, Pydantic, pytest                                                                               |
| ML layer        | Calibrated monotonic XGBoost champion with native exact TreeSHAP; calibrated logistic fallback                  |
| Frontend        | Accessible, deep-linkable HTML / CSS / JS cockpit with no build step                                            |
| Deployment & CI | Non-root Docker runtime, readiness probe, Render Blueprint, two-job GitHub Actions                              |
| Stage 2 targets | Supply IDBI sandbox records to the shipped outcome/temporal gates; add SSO/KMS and durable shared audit storage |



## **Near-zero to prototype. Scoped jointly for Stage 2.**

The prototype runs entirely on an open-source stack and free-tier hosting. Stage 2 production cost scales with transaction volume and is best scoped jointly with IDBI during the sandbox mentorship phase, once real data volume is known.

## The live decision stage.

Captured from the verified submission build: a calm NTC reversal stage beside a permanent review packet with file-divider tabs — Decision, Evidence, Model, Governance, Proof, and Sources — so nothing judges need is hidden behind a modal.

The screenshot displays the SaakhScore interface. At the top, a header bar includes the SaakhScore logo, a description of the MSME credit review process, and a 'Borrower file' dropdown menu currently showing 'Shree Ganesh Textiles (New-to-Credit)'. Below this, a grid of five borrower cards is visible: Shree Ganesh Textiles (New-to-Credit), Patel Wholesale Traders, City Corner Retail, Sunrise Auto Parts, and Himalayan Handicrafts. Each card lists the borrower's name, location, and credit grade with amount. The Shree Ganesh Textiles card is highlighted with a purple border. Below the grid, a 'Decision packet' section is open, showing a 'Five-pillar health ledger' with scores for Liquidity (16/20), Discipline (17/20), and Momentum (14/20). The interface is clean and modern, with a light beige background and purple accents.

**SaakhScore**  
MSME credit review, without hiding the audit trail.  
[Live API](#) / [xgboost pd proxy v1 model](#) / [Public synthetic cohort with sandbox feed contracts](#)

Borrower file  
Shree Ganesh Textiles (New-to-Credit)

| Borrower                              | Location               | Grade   | Amount     |
|---------------------------------------|------------------------|---------|------------|
| Shree Ganesh Textiles (New-to-Credit) | Textiles / Surat       | A grade | ₹27,00,000 |
| Patel Wholesale Traders               | Wholesale / Ahmedabad  | B grade | ₹32,80,000 |
| City Corner Retail                    | Retail / Nagpur        | D grade | ₹1,53,000  |
| Sunrise Auto Parts                    | Auto Components / Pune | C grade | ₹7,75,000  |
| Himalayan Handicrafts                 | Handicrafts / Dehradun | B grade | ₹1,20,000  |

SA-ntc-hero Textiles / Surat New-to-Credit / New-to-Bank

## Bureau rejection reversed.

Shree Ganesh Textiles (New-to-Credit) is approved for ₹27,00,000 after consented cash-flow, GST, UPI, EPFO-style, and bureau signals are reviewed. Reasons, model evidence, fairness checks, and proof sit in the review packet alongside.

[Read decision packet](#) [See model evidence](#)

### Decision packet

DECISION EVIDENCE MODEL GOVERNANCE PROOF SOURCES

#### Five-pillar health ledger

| Metric     | Score |
|------------|-------|
| Liquidity  | 16/20 |
| Discipline | 17/20 |
| Momentum   | 14/20 |

Model evidence tab: real-SBA benchmark splits with bootstrap intervals, DeLong-tested baselines, artifact integrity, and fairness slices.

Governance packet: audit count, validation metrics, fail-closed pilot gates, pilot KPIs, fairness slices, and controls.

Proof tab: truth boundary, judging rubric, competitor gap map, runbook, and API catalog.

## Verified, not just claimed.

**162/162**

Automated tests passing

**0.962**

Real-SBA out-of-time AUC

**0.926**

Recession-stress AUC

**418k**

Real loans, real charge-offs

The v2 benchmark (GET /model/sme-benchmark) is selected across 418,947 resolved real SBA 7(a) small-business loans at natural base rates (197,716-loan train split) and holds AUC 0.9623 / KS 0.82 on a genuinely later-in-time window (FY2017-19) and 0.9255 through a 257k-loan recession stress cohort — real outcomes, real time axis, with the whole experiment registry committed. The consumer-proxy holdout (AUC 0.7497 on 4,500 rows) remains as the conduct-pillar baseline. US proxy domain, disclosed; not an IDBI calibration.

# Stage 2 gates are live. Bank data is next.

## Phase 1

### Sandbox data intake

Connect authenticated feeds, profile distributions, and validate dated outcomes through /sandbox/pilot-readiness.

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## Phase 2

### IDBI-calibrated champion

Retrain on mature chronological cohorts and promote only when the shipped deployment gate confirms true OOT evidence.

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## Phase 3

### Fairness & monitoring

Run the implemented fairness gap, PSI, and reason-code stability checks on production-scale data.

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## Phase 4

### Bedrock-powered memos

Enable the optional AWS Bedrock provider in the pilot environment, with deterministic fallback.

## 13 LINKS

# Where to find everything.

→ GitHub Public Repository      <https://github.com/bansalbhunesh/id>

→ Animated Walkthrough      [github.com/bansalbhunesh/id/blob/main/docs/demo.gif](https://github.com/bansalbhunesh/id/blob/main/docs/demo.gif)

→ Final Product Link      <https://id-ysm9.onrender.com>

→ Demo Video (YouTube)      <https://youtu.be/0I5BY6DKB1E>